Activity 2 – Concept Note

1. Compare OSI Model and TCP/IP Model.
2. Network Troubleshooting:
   In a network, if the following network issues are happening
   a. Website not opening
   b. Cable unplugged
   c. Wrong IP address
   d. DNS failure
   e. Switch failure
   f. Firewall blocking traffic

   Mention the associated OSI Layer and Solution to troubleshoot the issue.

Network Issue	OSI Layer	Solution
Website not opening		
Cable unplugged		
Wrong IP address		
DNS failure		
Switch failure		
Firewall blocking traffic		

3. A smart temperature sensor sends temperature readings every 10 seconds to a cloud server. Occasionally, one reading is lost, but the next reading arrives successfully.
   a. Which transport protocol is more appropriate?
   b. Why?
   c. Which TCP/IP layer handles the sensor application?
   d. Which layer routes packets to the cloud?

4. A customer places an order on an e-commerce website. The order details are prepared, divided into smaller units, routed through different networks, and finally transmitted over a Wi-Fi connection.
   a. Which TCP/IP layer prepares the order information for communication?
   b. Which layer ensures the data is divided into smaller segments?
   c. Which layer determines the path to the destination?
   d. Which layer transmits the data over the wireless medium?